

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int bt[20], p[20], wt[20], tat[20];
6      int i, j, n, total = 0, pos, temp;
7      float avg_wt, avg_tat;
8
9      printf("Enter number of process:");
10     scanf("%d",&n);
11
12     printf("\nEnter Burst Time:");
13     for(i=0;i<n;i++)
14     {
15         printf("np%d:",i+1);
16         scanf("%d",&bt[i]);
17         p[i]=i+1;
18     }
19
20     for(i=0;i<n;i++)
21     {
22         pos=i;
23         for(j=i+1;j<n;j++)
24         {
25             if(bt[j]<bt[pos])
26                 pos=j;
27         }
28
29         temp=bt[i];
30         bt[i]=bt[pos];
31         bt[pos]=temp;
32
33         temp=p[i];
34         p[i]=p[pos];
35         p[pos]=temp;
36     }
37
38     wt[0]=0;
39
40     for(i=1;i<n;i++)
41     {
42         wt[i]=0;
43         for(j=0;j<i;j++)
44             wt[i]+=bt[j];
45
46         total+=wt[i];
47     }
48
49     avg_wt=(float)total/n;
50     total=0;
51
52     printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time");
53
54     for(i=0;i<n;i++)
55     {
56         tat[i]=bt[i]+wt[i];
57         total+=tat[i];
58
59         printf("np%d\t\t %d\t\t %d\t\t %d",
60                p[i], bt[i], wt[i], tat[i]);
61     }
62
63     avg_tat=(float)total/n;
64
65     printf("\nnAverage Waiting Time=%f", avg_wt);
66     printf("\nnAverage Turnaround Time=%f", avg_tat);
67
68     return 0;
69 }

```

Enter number of process: 3

Enter Burst Time:

P1: 45

P2: 32

P3: 18

Process	Burst Time	Waiting Time	Turnaround Time
P1	45	0	45
P2	32	45	77
P3	18	77	95

Average Waiting Time = 40.666668

Average Turnaround Time = 72.333336

Process exited after 7.533 seconds with return value 0

Press any key to continue . . . |